XMUM Library
Book Holdings Analysis

Descriptive statistics and data-quality review

Library Data Talk | Mathematics

9 October 2026

Source: four files in the supplied BookAttributes archive
Unit of analysis: individual catalog item records

Data sources and counting method

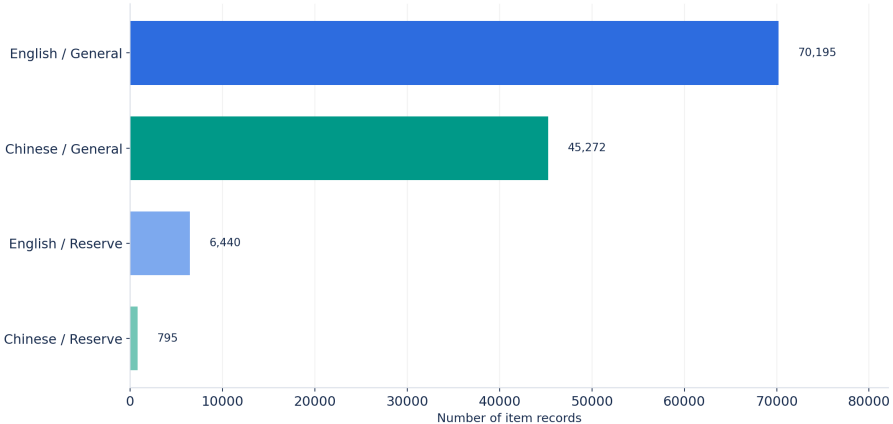
Supplied file	Item records
English General Books	70,195
English Course Reserve	6,440
Chinese General Books	45,272
Chinese Course Reserve	795
Total	**122,702**

Counting rule

122,702 item rows; 122,701 non-missing, unique barcodes. One item is missing its barcode. Distinct records do *not* imply distinct book titles.

Book-holdings dataset: four sources

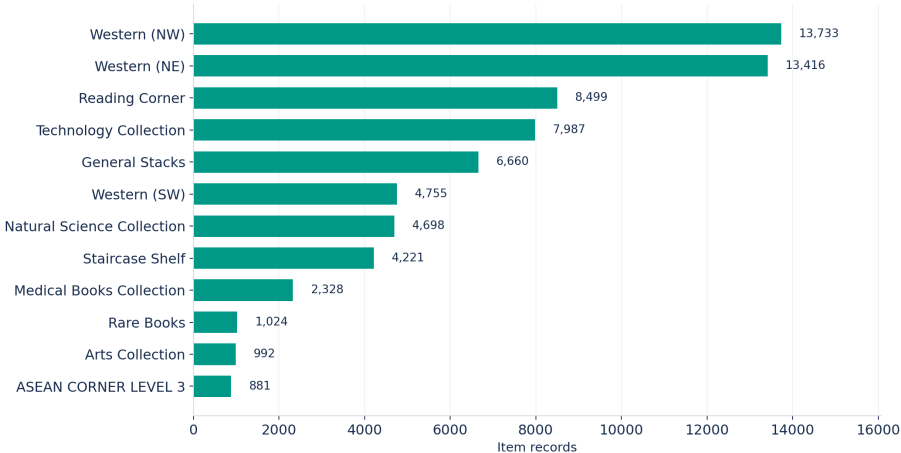
Record counts represent item rows, not distinct book titles



Total: 122,702 item records | 122,701 non-missing, unique barcodes

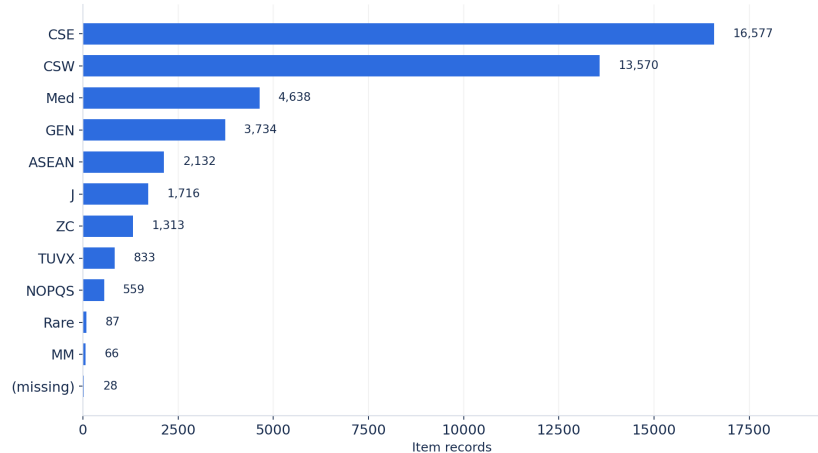
English general books: shelving locations

Top 12 locations; remaining locations not displayed



Chinese general books: location codes

Use actual "location" values; "Location Code" is a 13-row glossary, not per-item location



Location data: interpret the right field

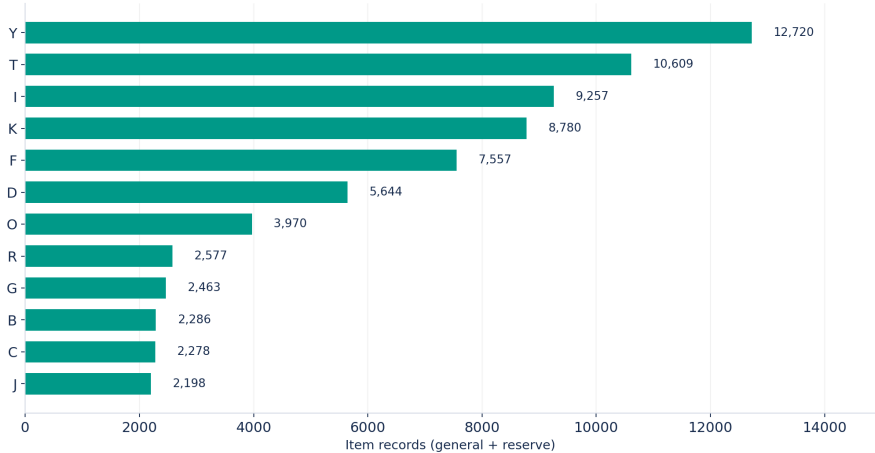
- Chinese General Books: `location` is the item-level location code.
- `Location Code` and `Unnamed: 9` contain a 13-row code glossary attached to the spreadsheet, **not** a per-book location assignment.
- Four codes are absent from that glossary: `ZC` (1,313), `TUVX` (833), `MM` (66), and `Tech` (4); 28 additional records have no location.

Interpretation

The 2,216 unlisted-code records need a verified codebook. They should not be called erroneous without confirmation.

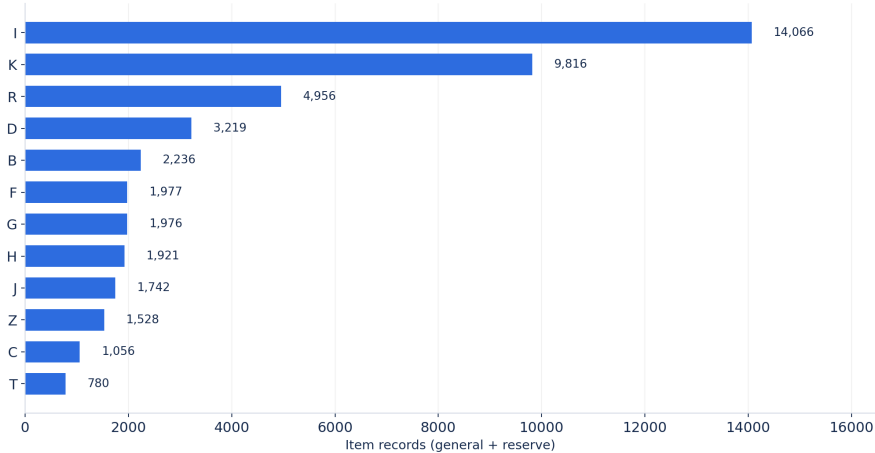
English holdings: leading call-number prefixes

First letter of the shelf call number; not a verified subject taxonomy



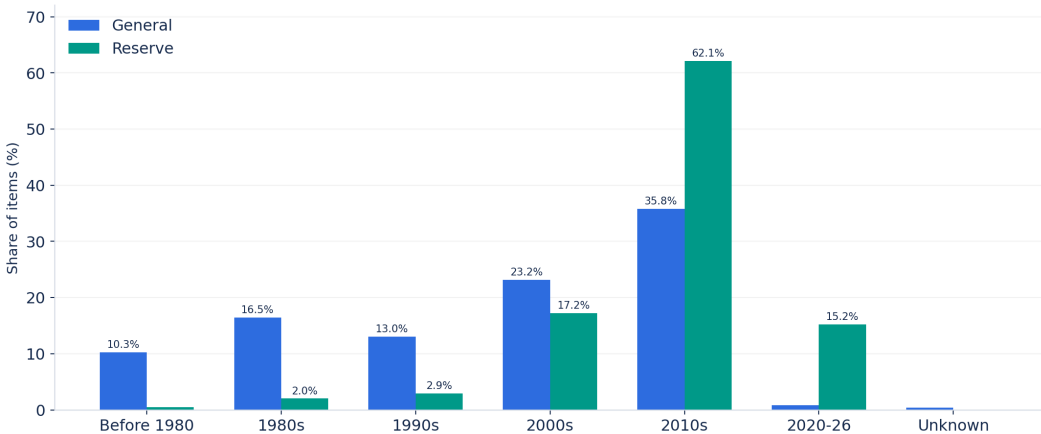
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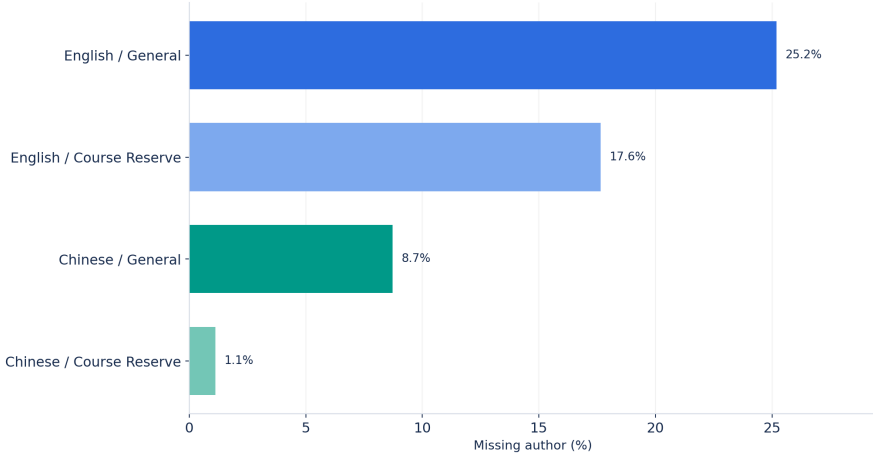
Chinese holdings: publication-year profile

Percent of items within each dataset; date is copyright/publication year, NOT acquisition year



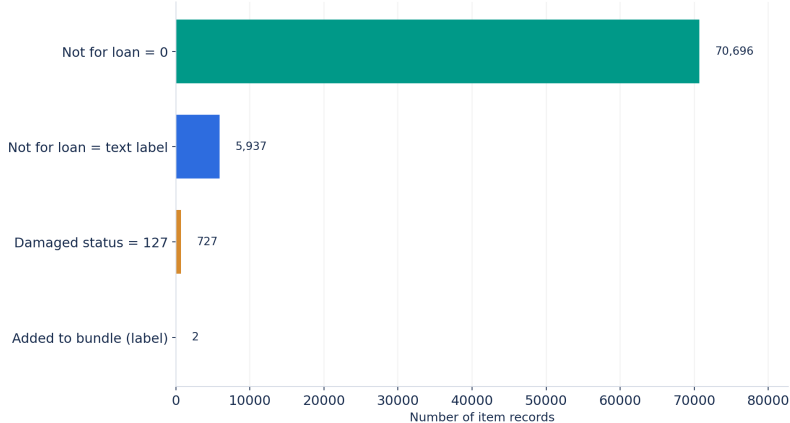
Metadata completeness: missing author field

Blank author is a metadata gap; some works may legitimately lack a named author



English holdings: raw library-status labels

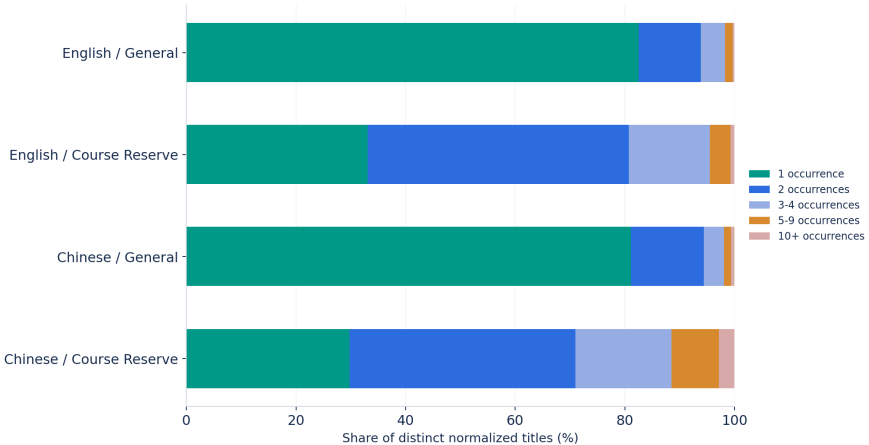
These are exported codes/labels; confirm Koha semantics before interpreting as loan/damage events



Categories are not mutually exclusive across the two different status fields.

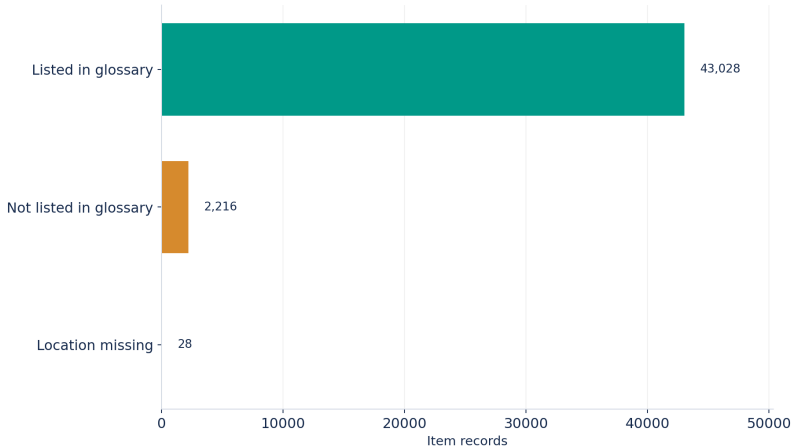
Repeated titles within each dataset

Distinct normalized title strings; repeats may be copies OR different volumes/editions



Chinese general books: location-code glossary coverage

Unlisted codes require documentation; they are not automatically wrong



Main findings

1. The dataset contains **122,702** item rows; English records account for **62.5%**, and General Books for **94.1%**.
2. The leading shelving location is **Western (NW)** for English General Books (13,733); the leading Chinese code is **CSE** (16,577).
3. The 2010s account for **35.8%** of Chinese General and **62.1%** of Chinese Reserve publication dates. These are *not acquisition trends*.
4. Missing author metadata is most prominent in **English General Books (25.2%)**.

Data limitations and recommended next steps

- **Validate catalog codes**: obtain the complete Chinese location glossary and confirm English Koha status values (especially 127).
- **Check anomalous fields**: one missing barcode, one Chinese publication-year value 7000, and 168 Chinese General records with no extractable year.
- **Keep title counts separate**: same titles may represent copies, multi-volume series, or editions; barcodes are the robust item identifier.
- **Do not infer borrowing demand**: item attributes alone cannot measure circulation or visits. Join borrowing and walk-in datasets for usage-based conclusions.

Reproducibility

All charts are generated from the untouched source files. Exported summary CSV tables and the Python script are supplied separately.